Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahaj-bhadja sahaj-bhadja

Summary

Software engineer with 2+ years of experience building **scalable cloud applications** across **Java**, **Python**, **JavaScript/TypeScript**, and **SQL**. Strong in object-oriented design, backend and full-stack feature delivery, automated testing, CI/CD, and AI-assisted engineering for stable, high-usage products.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University **Gandhinagar, In**
Bachelor of Technology in Computer Science and Engineering *November 2020 – May 2*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant cloud platform across **Azure SQL**, **PostgreSQL**, and cloud storage for 500+ users, reducing dashboard latency from **1.2s to 350ms** through schema, indexing, and query optimization.
- Built **FastAPI** data-integration services and release-backed validation workflows for **5K+ monthly jobs**, adding reconciliation checks and telemetry-style monitoring to sustain **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and shipped a **Flask + React** grading platform serving 150+ instructors, with modular REST workflows, user-facing dashboards, and **MySQL/AWS RDS**-backed services handling **10K+ submissions**.
- Implemented concurrent backend processing and **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while reducing grading turnaround time by **60%** and keeping production response stable under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **20+ unit/integration suites** using **Jenkins**, PyTest, and Mockito, contributing production code and test coverage across **3 operating systems** in an Agile Scrum environment.
- Optimized API and ETL components using **OOP patterns**, **AWS Glue**, and S3, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** platform with **REST APIs**, **WebSockets**, and microservices, supporting **8+ business workflows** and **2M+ synchronized records** for a user-facing operations product.
- Automated deployments and validation with **Jenkins**, Docker, and Azure SQL-backed services, reducing deployment time by **50%** while reaching **95%+ successful build validation**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built backend APIs and full-stack booking workflows with **React**, **TypeScript**, Flask, and Firebase across **50K+ records**, covering reservations, pricing, payment status, and real-time sync.
- Integrated webhook-driven payment flows and 12+ backend integration points, improving confirmation speed by **40%**, reducing manual payment handling by **70%**, and strengthening end-to-end system reliability.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, OOP, Client-Server Architecture
Frontend: React.js, Next.js, HTML, CSS, TailwindCSS
Databases: PostgreSQL, MySQL, SQL Server, MongoDB, Azure SQL, Relational Data Modeling, Query Optimization
Cloud & Infrastructure: AWS (RDS, S3, Glue), Microsoft Azure, Docker, Kubernetes, Jenkins, GitHub Actions
AI/ML: LLM Systems, GPT-4o, Claude, AI Coding Assistants, Prompt-Oriented Development
Engineering Practices: CI/CD, Unit & Integration Testing, Microservices, ETL Pipelines, Agile, Telemetry & Metrics, System Design

Projects

Chat2SQL | *Java, TypeScript, SQL, Docker*
- Built an AI-powered SQL workspace on top of Chat2DB, supporting **10+ databases** and accelerating query authoring, schema exploration, and structured execution across a **Java + TypeScript** codebase.

Big-Brain | *LLaMA-2, FAISS, PostgreSQL, REST APIs*
- Built a cloud-based AI document platform over **50K+ documents** with semantic search, summarization, and optimized retrieval pipelines, reducing latency by **40%** and supporting 200+ users.

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a fintech platform supporting **50K+ users** and **1M+ transactions** with secure bank linking, real-time synchronization, and production-style deployment workflows for high-volume product use cases.